An exact finite Gram cone for quadratic half-line selectors

Fix the published canonical source $c = (d, \lambda, n, m, C, A, D, \alpha, \beta) \geq 0$. Let $P(v), N(v), E(v), K(v)$ be the four symmetric 2×2 matrices of the canonical-completion note. The cone considered here consists precisely of sources for which some $J(v) = J_0 + vJ_1 + v^2 J_2$, with unrestricted real 2×2 coefficients, makes

$$G_J(v) = \begin{pmatrix} P+K & J \\ J^T & N+K \end{pmatrix}, \qquad H_J(v) = \begin{pmatrix} P+E & K-E-J \\ K-E-J^T & N+E \end{pmatrix} \succeq 0 \quad (v \geq 0). \tag{1}$$

Equality of this cone with all valid canonical sources is not assumed.

Lemma 1 (Quadratic matrix polynomials on a half-line)**.** *For real symmetric 4×4 matrices M_0, M_1, M_2, the polynomial $M(v) = M_0 + vM_1 + v^2 M_2$ is positive semidefinite for all $v \geq 0$ if and only if there are an 8×8 Gram matrix $S \succeq 0$ and a 4×4 Gram matrix $T \succeq 0$ with*

$$S = \begin{pmatrix} M_0 & B \\ B^T & M_2 \end{pmatrix}, \qquad T = M_1 - B - B^T. \tag{2}$$

Here B is an unrestricted real 4×4 matrix.

Proof. The reverse implication is the identity $M(v) = [I \ vI] S [I \ vI]^T + vT$. Conversely $M(t^2)$ is a real matrix polynomial of degree at most four which is positive semidefinite on the real line. The real-line matrix Fejér–Riesz theorem (Zalar, *Matrix Fejér–Riesz theorem with gaps*, Theorem 1.2) gives a polynomial matrix factor of degree at most two; taking real and imaginary parts gives real matrix squares. Averaging their values at t and $-t$ yields

$$M(v) = (R_0 + vR_2)^T (R_0 + vR_2) + vR_1^T R_1.$$

Set $S = [R_0 \ R_2]^T [R_0 \ R_2]$ and $T = R_1^T R_1$. The degree bounds are sharp: $M(v) = vI$ needs the half-line term, and $M(v) = v^2 I$ needs a linear factor. No strict positivity, invertibility, or nonzero leading coefficient is used. The result also holds over real closed fields by transfer of this finite first-order assertion. □

Here are the evaluated integer coefficient formulas. Write $D(a, b) = \operatorname{diag}(a, b)$ and $O(a) = \left(\begin{smallmatrix} 0 & a \\ a & 0 \end{smallmatrix}\right)$. In the coefficient order $1, v, v^2$,

$$\begin{aligned}
(P_0, P_1, P_2) &= (D(0, 4m + 4\beta + 2d), O(-4m), D(4m, 4\beta + 2d)), \\
(N_0, N_1, N_2) &= (D(4C + 4D + 2d, 0), O(4D + 2d), D(4C, 4D + 2d)), \\
(E_0, E_1, E_2) &= (D(4A, 4\alpha), O(-4\alpha - 2d), D(4A + 4\alpha + 2d, -2d)), \\
(K_0, K_1, K_2) &= (D(4\lambda, 4n), O(4\lambda), D(0, 4\lambda + 4n)).
\end{aligned}$$

For $j = 0, 1, 2$ define

$$G_j = \begin{pmatrix} P_j + K_j & J_j \\ J_j^T & N_j + K_j \end{pmatrix}, \qquad H_j = \begin{pmatrix} P_j + E_j & K_j - E_j - J_j \\ K_j - E_j - J_j^T & N_j + E_j \end{pmatrix}. \tag{3}$$

These are affine linear with integer coefficients in the source and selector entries. In addition to the twelve selector entries, introduce unrestricted 4×4 matrices B_G, B_H. The complete coefficient elimination is

$$\boxed{\begin{aligned}
&S_G = \begin{pmatrix} G_0 & B_G \\ B_G^T & G_2 \end{pmatrix} \succeq 0, \qquad S_H = \begin{pmatrix} H_0 & B_H \\ B_H^T & H_2 \end{pmatrix} \succeq 0, \\
&T_G = G_1 - B_G - B_G^T \succeq 0, \quad T_H = H_1 - B_H - B_H^T \succeq 0.
\end{aligned}} \tag{4}$$

Proposition 2. *Projection of (4) onto $c \geq 0$ is exactly the cone in (1), including every zero or degenerate source face. It has four Gram blocks of orders $8, 8, 4, 4$, total order 24, and $12 + 16 + 16 = 44$ auxiliary scalars, with no remaining coefficient equations. Including the nine scalar blocks $[c_i]$ gives an ambient affine semidefinite representation of total order 33.*

Proof. If (4) holds, (2) reconstructs each matrix in (1) for every $v \geq 0$. If (1) holds, apply the lemma independently to its two quadratic matrices; their off-diagonal Gram blocks give B_G, B_H in (4). Thus no selector has been discarded and no extra source condition introduced. All operations are coefficient identities without division, so zeros and singular Gram matrices remain literal cases. In particular the all-zero data are feasible. □

For comparison, retaining all Gram entries would give $72 + 20 + 12 = 104$ unknowns and 60 scalar coefficient equations. Formula (4) solves these equations explicitly, leaving 44 unknowns; it makes no symmetry assumption on either B or any J_j.

Lemma 3 (Bounded auxiliaries and closed projection)**.** *Put $s = \sum_i c_i$. Every feasible auxiliary vector satisfies $\|y\|_\infty \le 16s$, and the projected source cone is closed. More precisely,*

$$g := \operatorname{tr} S_G = 8d + 16\lambda + 16n + 8m + 8C + 8D + 8\beta,$$
$$h := \operatorname{tr} S_H = 8d + 8m + 8C + 8D + 8\beta + 16A + 16\alpha, \qquad \operatorname{tr} T_G \le g, \quad \operatorname{tr} T_H \le h.$$

Proof. All diagonals of S_G, S_H depend only on c, and $g, h \le 16s$. Their PSD principal minors bound every B_G, B_H, J_0, J_2 entry by $g/2$ or $h/2$. Put $a_i = (S_G)_{ii} + (S_G)_{i+4,i+4}$, $0 \le i < 4$. Since $(G_1)_{ii} = 0$, positivity gives $0 \le (T_G)_{ii} = -2(B_G)_{ii} \le a_i$; summing gives the trace bound, and likewise for H. For each selector entry $(J_1)_{ij}$, its position in G_1 has distinct indices $r = i$, $t = j + 2$. The PSD entry bounds give $|(J_1)_{ij}| \le |(T_G)_{rt}| + |(B_G)_{rt}| + |(B_G)_{tr}| \le a_r + a_t \le g$. This bounds all 44 auxiliaries. Along a convergent sequence of feasible sources the auxiliaries therefore have a convergent subsequence, whose limit remains feasible by closedness of the PSD cones. With arbitrary signed source inputs the same proof uses $16 \sum_i |c_i|$, so the order-24 projection is closed as well. Moreover fixed-source feasibility cannot be weakly infeasible. Otherwise a sequence approaching the PSD cones would either have bounded auxiliaries, giving a feasible limit, or an unbounded subsequence. Dividing the latter by its auxiliary norm and taking a subsequence would give feasible homogeneous data with source zero and a nonzero auxiliary vector, contradicting the bound. □

For each feasible source, put $T_2 = \operatorname{diag}(1, -1)$. The published matrices obey $M(-v) = T_2 M(v) T_2$. Define the selector on $v < 0$ by $J_-(v) = T_2 J_+(-v) T_2$. Congruence by $\operatorname{diag}(T_2, T_2)$ carries both inequalities from $-v > 0$ to $v < 0$. Thus (4) implies canonical validity by the pointwise completion theorem; finite axes and projective endpoints follow from the full homogeneous parent. This argument asserts only this inclusion. It does not prove that every valid source admits a quadratic half-line selector. The reflected selector is generally piecewise polynomial, so the global polynomial-selector counterexample does not contradict this construction.

The two accompanying COO files evaluate (3)–(4) exactly. They use all 53 ordered parameters (nine source and 44 auxiliary entries), full symmetric COO indexing, and respectively omit or include the external source-sign blocks. The standard-library checker reconstructs both original matrix quadratics for every parameter basis vector at three rational testing values; hence it checks every coefficient. It performs no solver call, numerical search, or universal-adequacy test.